

Part B Question 6

Algorithm:- we take the inputs from user and calculate the cost according to rules and apply discount if the person ^{has} membership. Checks for the priority and change the vehicle if conditions get true.

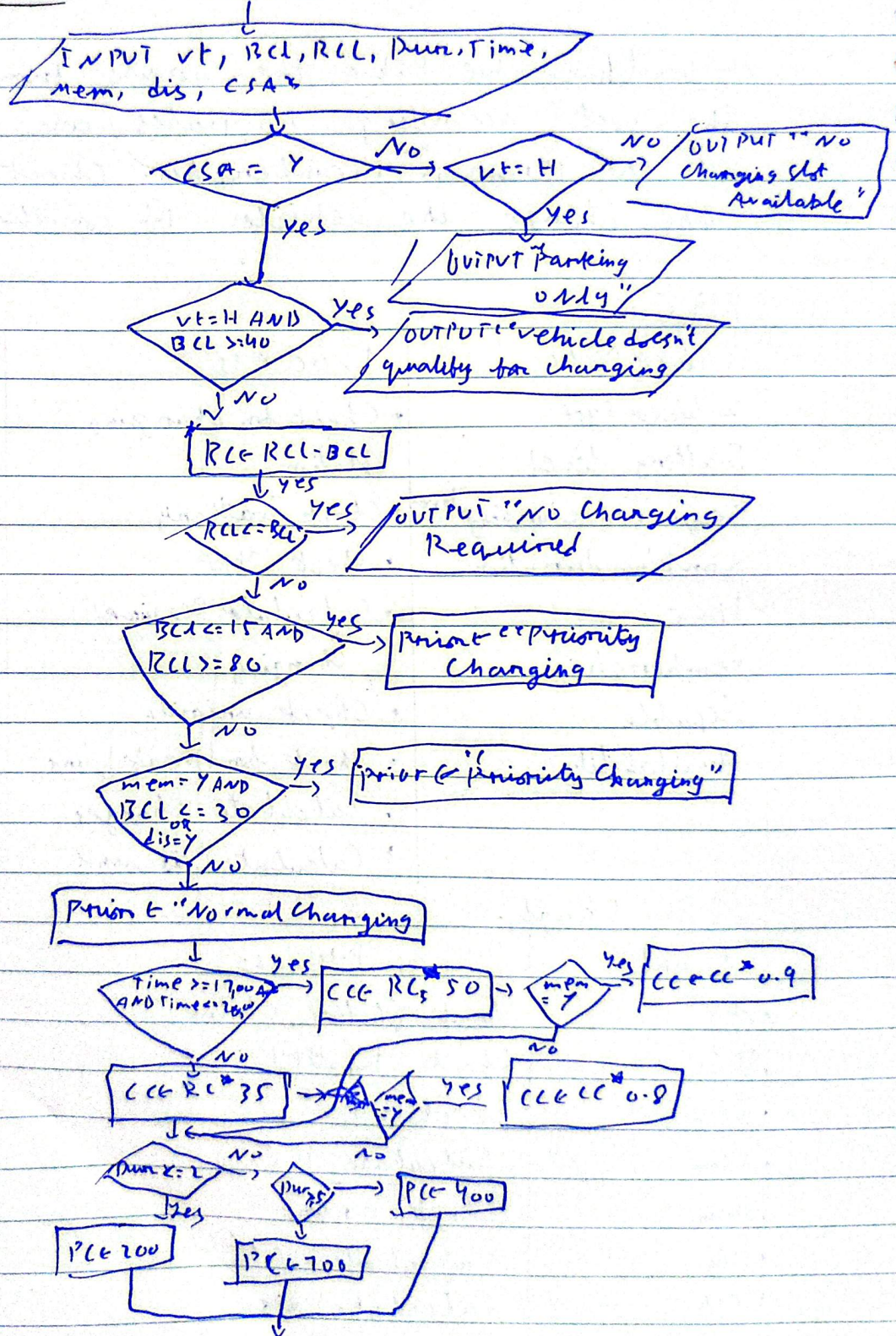
PAC:-

Given Data	Process	Required Results
Vehicle type	• Check for changing station	• vehicle type
Battery level	• Check battery	• Current battery
Required charging level	• Check type	• Required charging
parking duration	• Calculate Required Changing	• Priority
Time	• Check priority	• Status
membership	• Check for peak hours	• Cost
Disable	• Calculate charges	• Discount
Availability	• Calculate discount	• Final amount
		• warning message

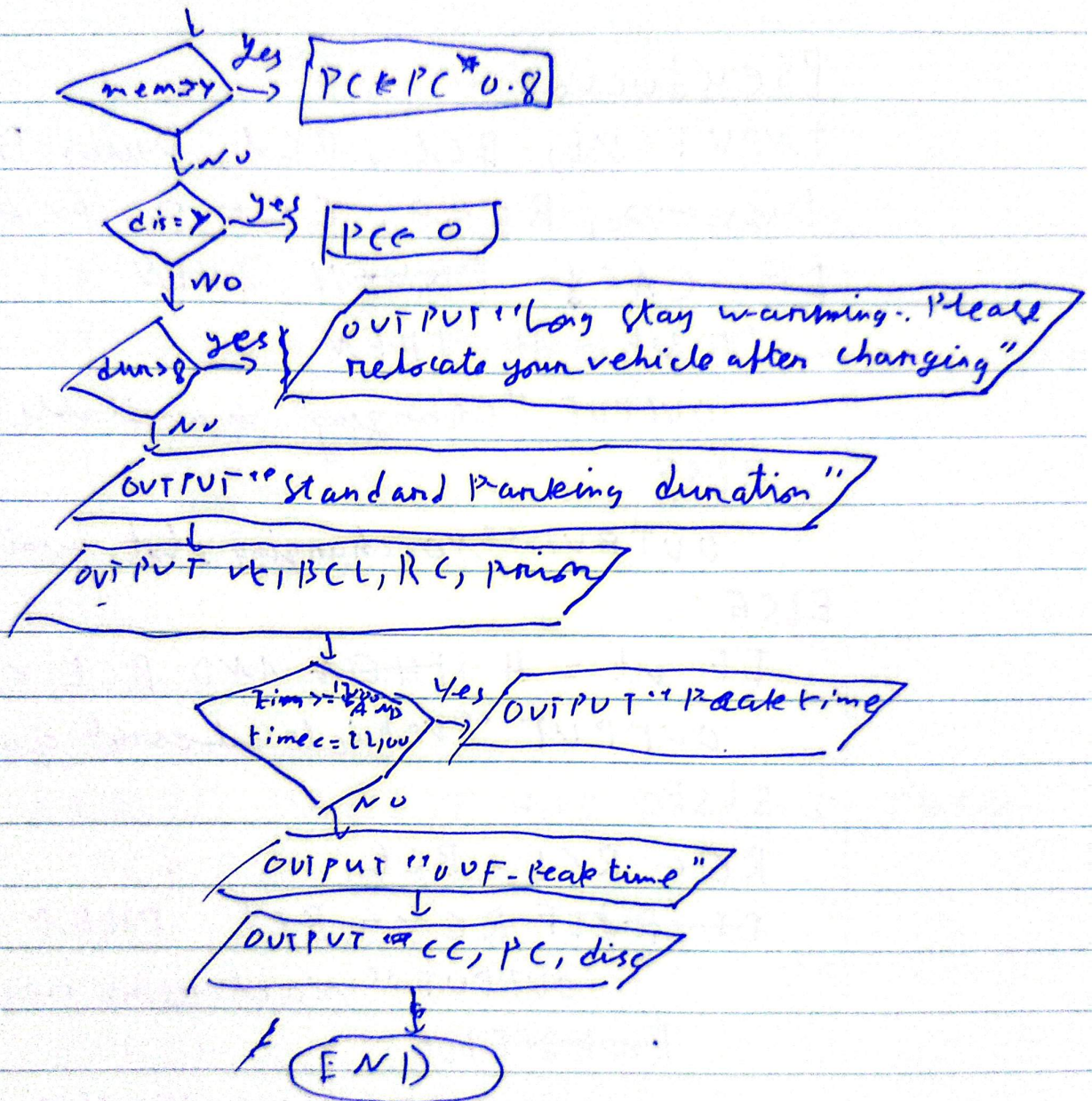
IPO Chart:-

Input	Process	Output
• vt	Check station status	• vt
• BCL	Check for BCL	• BCL
• RCL	Check vt	• RC
• Time	Calculate RC	• CP
• mem	Check p-nom	• PP status
• Dis	Calculate cost	• P cost
• CSA	Calculate disc	• Final Amount
		• message

Flow Chart :- Start



Date:



Pseudocode:-

```

INPUT vt, BCL, RCL, Dur, Time, Mem, Dis, CAS
Disc ← 0, RC ← 0, CC ← 0, PC ← 0, FC ← 0, pri ← ""
IF CAS = TRUE THEN
    IF vt = H THEN
        OUTPUT "Charging unavailable - parking only"
    ELSE
        OUTPUT "No charging slot available"
    ELSE
        IF vt = H THEN AND BCL < 40 THEN
            OUTPUT "Vehicle does not qualify for EV charging"
        ELSE
            RC ← RCL - BCL
            IF IF RC ≤ BCL THEN
                OUTPUT "No charging required"
            ENDIF ELSE
                IF IF BCL ≤ 15 AND RCL ≥ 80 THEN
                    pri ← "Emergency Charging Priority"
                ELSE IF Mem = "Y" AND BCL ≤ 30
                    pri ← "Priority Charging"
                ELSE
                    pri ← "Normal Charging"
                ENDIF
            IF Time ≥ 1700 AND Time ≤ 2200 THEN
                OUTPUT "peak time"
                CC ← 35 Dur 50 * Dur
                IF Mem = Y THEN AND NOT Emergency THEN
                    Disc ← 10
                
```

```

CC ← CC * 0.9
ENDIF
ELSE
CC ← 35 * Dur → OUTPUT "OFF Peak time"
IF mem = "Y" THEN AND NOT Emergency THEN
    Disc ← 20
    CC ← CC * 0.8
ENDIF
ENDIF
IF Dur ≤ 2 THEN
    PC ← 200
ELSEIF Dur ≥ 2 AND Dur ≤ 5 THEN
    PC ← 500
ELSE
    PC ← 700
ENDIF
IF mem = Y THEN
    PC ← PC * 0.8
ENDIF
IF Dis = Y THEN
    PC ← 0
ENDIF
IF Dur > 8 THEN
    OUTPUT "Please Relocate your vehicle
    after charging"
ELSE
    OUTPUT "Standard parking duration"
ENDIF

```

ENDIF

ENDIF

ENDIF

OUTPUT "vehicle type is ", vt

OUTPUT "Current Battery percentage is ", BCL

OUTPUT "Required Charging percentage is ", RC

OUTPUT "Charging Priority is ", prprio

OUTPUT "Charging Cost is ", CC

OUTPUT "Parking cost is ", PC

OUTPUT "Discount is ", Disc

OUTPUT "Final Payable Amount is ", CC + PC

~~OUTPUT~~